

# KIMI INDEPENDENT REVIEW VERDICT — LPW-CAND-20260912 (local-path candidate, existential cubic lower bound)

Reviewer: KIMI line (lead coordinator + four independent interface reviewers RA/RB/RC/RD).  
Date: 2026-09-13. Target: 02\_LOCAL\_PATH\_LOWER\_BOUND\_CANDIDATE.md v1.0, body sha256 cf58f72eb0399c626160c143b50f674ff3bd5c449225211edd6fd378a374e1a5 (matches the package MANIFEST). Review packet: 03\_INDEPENDENT\_REVIEW\_PACKET.md (six interfaces, separate dispositions required). Exposure disclosure: the full candidate and packet were read before review — ordinary hostile review, not blind reconstruction. Review lineages: five (lead + RA + RB + RC + RD), each with its own independent scripts; no author-script lineage consumed.

**VERDICT: APPROVE — all six interfaces PASS WITHIN SCOPE at the exact file version.**

The candidate’s theorem LPW is endorsed: **there exist  $c > 0$  and  $r > 0$  such that  $1 - q(r, 6/5) c \cdot r^3$  for all  $0 < r \leq r$** , under the exact normalized periodized side-24 two-dimensional Bargmann–Fock field and the exact six-pin typed maximum–saddle pair-Palm law (pins  $f(M_r) = 6/5$ ,  $f(S_r) = 6/5 - r^3/6$ ,  $f = 0$  at  $M_r = (-r/2, 0)$ ,  $S_r = (r/2, 0)$ ; weight  $W_r = |\det H_M \det H_S| \cdot 1\{H_M \geq 0, \det H_S < 0\}$ ;  $q(r, b) = P_r\{D_f(M_r) = S_r\}$ , no adjacency conditioning,  $r$  untouched).

## Interface dispositions (separate, per the packet’s requirement)

| interface                             | reviewer | verdict                                       | file (sha256)           |
|---------------------------------------|----------|-----------------------------------------------|-------------------------|
| R1 law and topology                   | RA       | PASS WITHIN SCOPE                             | RA_R1_R6.md 3b7cda65... |
| R2 six-/ten-coordinate Gaussian limit | RB       | PASS WITHIN SCOPE                             | RB_R2.md 07ceb63c...    |
| R3 conditional C norm control         | RC       | PASS WITHIN SCOPE (2 non-blocking amendments) | RC_R3.md 29dcbc23...    |
| R4 deterministic lift                 | RD       | PASS WITHIN SCOPE                             | RD_R4_R5.md 09aaac1d... |
| R5 normalizer and final powers        | RD       | PASS WITHIN SCOPE                             | RD_R4_R5.md 09aaac1d... |
| R6 scope and status                   | RA       | PASS WITHIN SCOPE                             | RA_R1_R6.md 3b7cda65... |
| lead’s independent evidence pack      | lead     | consistent with all six                       | LEAD_EVIDENCE_PACK.md   |

## What was independently verified (each by at least two independent lineages)

- The exact witness algebra:**  $F^*$ ’s pinned maximum/saddle  $(\det 6, -14)$ , the path height  $R(x) = (2x+1)^2(12x^2+8x-17)/240$ , the exact identity  $R + 99/1280 = (4x+5)^2(48x^2-40x+1)/3840$ ,  $\min R = -99/1280 > -1/6$ , endpoint  $9/16 > 0$ , clearance  $103/3840 > 0$  under  $\epsilon = 1/16$ , Hessian interval bounds  $(81/16, -1673/128)$ , weight product 65.
- The topological implication (R1):** a strict local maximum at  $b$  with a path to a point above  $b$  whose minimum exceeds  $f(S)$  cannot die at  $S$  — proved tie-free; all four adversarial escapes fail; the

needed assumptions (a.s. Morse + distinct critical values, measurability) are already the estimand's definitional package; NO Morse–Smale is consumed.

3. **The Hermite normalization** (R2/lead):  $V$  row-major with  $\det -1$ ; the displayed  $T_r$  with  $\det -r^{\wedge}\{-5\}$ , invertible for all  $r > 0$ ;  $u_r = (b - r^3/12, -r^2/4, 0, 0, 0, 1/3)$ ;  $U_r \rightarrow U_0$  in Gaussian  $L^2$  with covariance convergence (exact through cubics; trapezoidal cancellations exact).
4. **The ten-jet nondegeneracy** (R2): the spectral-support argument is sound, and numerically the exact field's ten-jet covariance at 0 has certified  $\_min = 0.126553449667289$  (eigen-residual  $6.9e-50$ , spectral tail  $1.2e-185$ ); the  $r = 0$  Schur complement is  $\text{diag}(2, 1/2, 1/2, 1/6)$  at machine precision; rungs  $r = 0.5...0.025$  converge to the endpoint ( $\_min \rightarrow 0.12656$ ); conditional means  $\rightarrow (-b, 0, 0, 0)$ .
5. **The density lower bound** (R2): the argument chain (uniform PD Schur + bounded determinants/inverse/means on the compact parameter set  $[0, r\_G] \times K\_J$ ) is sound; evidence at  $r = 0.025$ : density of  $J$  at the thin-box center  $1.2832e-3$ , minimum over 16 corners  $1.2781e-3$ .
6. **The conditional C moment** (R3): all six attack links held — the regression version is everywhere-defined (no version trap), the representers'  $C$  norms are uniform (Cauchy–Schwarz + covariance convergence, no hidden  $r$ -blowup through the Hermite frame), the residual's moments are finite and  $v$ -independent,  $\Gamma_r^{\wedge}\{-1\}$  is bounded on  $[0, r\_G]$ , the Markov  $1/2$  bound is pointwise in  $j$ , and the jet-box integration is measure-theoretically clean (no independence asserted, no subtraction-order error).
7. **The Taylor lift** (RD/lead): all six 1D pin identities exact ( $1/24, 1/48, 5/16, 1/128, 23/384$ , and the  $y$ -equation  $Kr/48$ ); the 2D remainder  $(9/2)Kr$ ; monomial norms  $1,2,1,4,2,12$ ; total  $2089/384 < 8$  (reproduced term-by-term); basis norms  $4,4,2,6$ ; tolerance  $3/64 < 1/16$ ; Hessian scaling  $r$  per matrix,  $r^2$  per determinant,  $r$  per pair.
8. **The normalizer and power ledger** (RD):  $Z_r - 4B_3r^2 = C_Z r^2$  with correct polarity;  $Z_r > 0$  grounded by the constructed event (no circularity);  $E_r - K_J$  for  $r \leq 1$ ; volume  $32 r$ ; numerator  $1040m - r$ ; quotient  $c = 1040m - /C_Z = 260m - /B_3$ ; exponent  $1 + 4 - 2 = 3$ .
9. **The estimand** (R1/lead): the candidate's  $q$  is exactly the program's  $p_r$  (W10 §1 firewall; manuscript Definition 2.8's “window” phrasing resolves to the exact pin  $f(S) = b - r^3/6$ ; the corrected pin basis, the raw six pins, and  $T_r$  span the same  $-$ algebra; identical weight, types, RN structure; no adjacency conditioning).
10. **Scope discipline** (R6): all constants structurally  $r$ -independent; no sampled finite-difference modulus; no planar-moment substitution; no silent promotion (GP-LB-STAT-004 unchanged; the accepted 2D lower campaign stays OPEN); conclusion strictly existential (no  $\liminf$  coefficient, no numerical  $c$  or  $r$ , no dimension-3 or adjacency variant;  $\_r$  retained).

## Amendments and scoped residues (none blocking)

- A1 (RC): the residual field is nonstationary, so its  $C$ -moment bound should be stated explicitly via the triangle inequality through the representers (substance present; presentation gap).
- A2 (RC): the  $M$  index convention ( $|| = 4$  vs  $|| = 4$ ) should be stated once (immaterial to §8).
- S1 (RA, non-blocking): equality with GP-DER-118-v1.10 §0 (candidate line 48) is CANNOT-VERIFY-locally — that body is absent from the local corpus (Drive ID 1qc6ep2S4PIoPMEWDsdDQI1dr0LOWJiK9; hashes c1d6e559.../9b7901e1... as reported by the package). The estimand of record is local and matches independently of that body.

## Consequences for the program (stated exactly, no promotion)

1. The existential cubic lower bound stands endorsed at review grade by an independent route that consumes NONE of the open quantitative obligations (WP,  $\Lambda/H$ -B3, exit, -LOC, Bonferroni, far-field, SARD-G,  $\_r$ ). This is the program's first endorsed lower theorem for the 2D problem at any scope:  $1 - q(r, 6/5) \leq c \cdot r^3$  for all sufficiently small  $r$ , some  $c > 0$ .
2. The candidate does NOT establish: the 0.9144-class constant, a liminf coefficient, a numerical  $c$  or  $r$ , a full lifetime distribution, a two-sided law at explicit constants, or any dimension-3 or adjacent-conditioned variant. The K3 assembly (K3-THM-001, conditional liminf at measured/mixed tier with hypotheses H1–H12) is a SEPARATE, logically independent statement at its own tiers; the candidate neither uses nor repairs its premises. The author-side state view's "K3-THM-001 REFUTED AS WRITTEN / NONCONTROLLING" attaches to the v1.1-lineage defects already quarantined by the K3 campaign; the K3 conditional assembly stands on its own receipts.
3. Follow-up available (not performed here): the constant expression  $c = 1040m / C\_Z$  is in principle evaluable (RB's endpoint density evidence  $m = 1.28e-3$ ;  $B\_3$  via the regression moments) — a certified evaluation would give a small but explicit numerical  $c$  and  $r$ .
4. The ratified upper chain (AO48-OPR-045) is untouched; combining it with this endorsed lower bound gives the two-sided  $\Theta(r^3)$  at the existential level, subject to the upper chain's own grades and to operator adjudication of any status change.
5. **Status firewall (exact):** this review endorses the candidate's mathematics at review grade. It changes no LS-CTL Boolean, eligibility predicate, theorem status, RP status, AO48 operator record, or q0 package status. Any status change requires separate operator adjudication. P0.1 remains HOLD; the accepted 2D lower campaign's quantitative obligations remain OPEN at their own grades.

## Falsifier (what would kill this endorsement)

A counterexample at any interface: a valid elder-rule interpretation in which  $M$  dies at  $S$  despite the path; a failure of the ten-jet covariance's positive-definiteness on the exact field; a conditional  $C$  moment that blows up uniformly over the compact box; a sign/typing error in the Taylor lift; a normalizer that is not  $O(r^2)$ ; or a proof that the program's  $q$  differs from the candidate's under the records of record. Any one retracts the corresponding interface's PASS and, if load-bearing, this endorsement.

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